1. Provide a definition/explanation for **any four of the following terms**. Be sure to clearly associate each term with its definition/explanation. (20%/20%)

   Algorithm, Control Structure, Data Type, Method, Object

2. Provide a complete and working Java program that reads three numbers as input, representing the dimensions of a box, then computes and prints the volume of that box. Consider these example interactions. (15%/35%)

```
c:\>java Straightforward1
3
4
5
Volume is 60

c:\>java Straightforward1
17
10
2
Volume is 340
```

3. Consider the following complete and valid Java source program. For each item in the table provide the line number(s) of a statement that presents an instance of that item. Note that multiple instances may be present and you are being asked to **identify just one instance of each item**. In addition to writing the line number in the table you are to also highlight (using a highlighter pen) the corresponding text in the program and annotate each with the corresponding letter. (15%/50%)

```
 1 import java.util.Scanner;
 2 public class Coins {
 3    public static final int CENTS_PER_DOLLAR = 100;
 4    public static final int CENTS_PER_HALF = 50;
 5    public static final int CENTS_PER_QUARTER = 25;
 6    public static final int CENTS_PER_DIME = 10;
 7    public static final int CENTS_PER_NICKLE = 5;
 8
 9    // Scanner object through which input data is read from the keyboard.
10    public static Scanner keyboard = new Scanner(System.in);
11
12    // Global input and output variables.
13    static int moneyInCents;  // (input) amount expressed in cents
14
15    public static void main(String[] args) {
16       do {
17          // Read user's response and store it in the input variable
18          moneyInCents = keyboard.nextInt();
19          // Report the amount as dollars and cents
20          System.out.println("Equivalent to:");
21          // Call to determine and report coin equivalence
22          determineCoins(moneyInCents);
23       } while (moneyInCents > 0);
24    }
25
26    public static void determineCoins(int moneyInCents) {
27       System.out.println("  "+fullDollars(moneyInCents)+" dollars,");
28       System.out.println("  "+fullResidualHalfs(moneyInCents)+" half-dollars,");
29       System.out.println("  "+fullResidualQuarters(moneyInCents)+" quarters,");
30       System.out.println("  "+fullResidualDimes(moneyInCents)+" dimes,");
31       System.out.println("  "+fullResidualNickels(moneyInCents)+" nickels,");
32       System.out.println("  "+fullResidualPennies(moneyInCents)+" pennies.");
33    }
34
35    public static int fullDollars(int totalCents) {
36       return totalCents / CENTS_PER_DOLLAR;
37    }
38
39    public static int fullResidualHalfs(int totalCents) {
40       return (totalCents % CENTS_PER_DOLLAR) / CENTS_PER_HALF;
41    }
42
43    public static int fullResidualQuarters(int totalCents) {
44       return (totalCents % CENTS_PER_HALF) / 25;
45    }
46
47    public static int fullResidualDimes(int totalCents) {
48       return (totalCents % 25) / 10;
49    }
50
51    public static int fullResidualNickels(int totalCents) {
52       return ((totalCents % 25) % 10) / 5;
53    }
54
55    public static int fullResidualPennies(int totalCents) {
56       return (totalCents % 5);
57    }
58
59 }
```

LABEL	Item	Line #s
A	Actual Parameter	
B	Arithmetic Expression	
C	Arithmetic Operator	
D	Comment	
E	Constant	
F	Formal Parameter	
G	Identifier	
H	Literal	
I	Loop	
J	Method	
K	Object	
L	Relational Expression	
M	Relational Operator	
N	Reserved Word	
O	Variable	

4. Consider the following correct and complete Java program. As your answer to this question you are to **provide the exact output produced by each of the "print statements" in the corresponding spaces provided**. This question is really about data types, operations and expressions. (15%/65%)

```
 1 public class EvaluateV1{
 2    public static String home = "LA Dodgers";
 3    public static String visitor = "NY Yankees";
 4    public static char c = ' ';
 5    public static void main (String[] args) {
 6       boolean b = true;
 7       double e;
 8       double w = 3.14;
 9       double what = 0.1;
10       int x = 17;
11       int y = 23;
12       int z;
13
14       z = x;                 System.out.println(z);
15       z = x + 3;             System.out.println(z);
16       z = (y % x);           System.out.println(z);
17       z = ((y / 10) * 10);   System.out.println(z);
18       e = w * 3;             System.out.println(e);
19       e = x + y;             System.out.println(e);
20       z = (x - (int)w);      System.out.println(z);
21                              System.out.println("$"+(x+y)+"$");
22                              System.out.println("$"+x+y+"$");
23       z = home.length();     System.out.println(z);
24       printSomething();
25       printOther(visitor);
26       if(x > y) {            System.out.println(f(home));
27       } else {               System.out.println(f(visitor));
28       }
29                              System.out.println(x >= z);
30                              System.out.println(b);
31    }
32
33    public static void printSomething() {
34       System.out.println(home);
35    }
36
37    public static void printOther(String s) {
38       int idx = s.indexOf(c);
39       String part = s.substring(0,idx);
40       System.out.println(part);
41    }
42
43    public static String f(String s) {
44       int m = s.length() / 2;
45       return s.substring(0,m);
46    }
47 }
```

5. As your answer to this question you are to provide the missing statement(s) that will complete the program so that it behaves as illustrated. A correct answer that makes appropriate use of the given **writeChars** method will earn substantially more credit that a correct answer that does not. (10%/75%)

```
Enter:>5
    *
   **
  ***
 ****
*****
```

```
Enter:>8
       *
      **
     ***
    ****
   *****
  ******
 *******
********
```

```
import java.util.Scanner;
public class Figure2 {
   static Scanner keyboard = new Scanner(System.in);

   private static int getIntFromUser(String prompt) {
      System.out.print(prompt);     // Prompt user to enter input value
      return keyboard.nextInt();    // Read and return user's response
   }

   public static void writeChars(char ch, int number) {
      for(int i = 1; i <= number; i++) {
         System.out.print(ch);
      }
   }

   public static void main(String[] args) {
      int measurement = getIntFromUser("Enter:>");

}
```

6. As your answer to this question you are to provide the missing statement(s) that complete the functional method, `middleNameOf`, that accepts a string value representing a person's "full name" (such as illustrated in the example) and returns the middle name portion of that name. (10%/85%)

```
c:\>java Names
Enter full name:>Paul Michael Jackowitz
Middle name is "Michael"

Enter full name:>Alex Gram Bell
Middle name is "Gram"

Enter full name:>Winnie da Pooh
Middle name is "da"

Enter full name:>Abe Lincoln
Middle name is ""

Enter full name:>
```

```
import java.util.Scanner;
public class Names {
   static Scanner keyboard = new Scanner(System.in);

   private static String getStringFromUser(String prompt) {
      System.out.print(prompt);   // Prompt user to enter input value
      return keyboard.nextLine(); // Read and return user's response
   }

   private static final char QUOTES = '"';
   private static String quoted(String s) {
      return QUOTES + s + QUOTES;
   }

   public static void main(String[] args) {
      String fullName;
      do {
         fullName = getStringFromUser("Enter full name:>").trim();
         if(fullName.length() > 0) {
            System.out.println("Middle name is " + quoted(middleNameOf(fullName)));
            System.out.println();
         }
      } while(fullName.length() > 0);
   }

   public static String middleNameOf(String name) {

   }
}
```

7. As your answer to this question you are to provide the missing statement(s) that complete the functional method, `hotterOf`, that accepts two double values, the first representing a temperature value in Fahrenheit and the second representing a temperature value in Celsius. The method returns either "`F`" or "`C`" as its value indicating which of the two measurements represents a higher temperature. In the event the two temperatures are equivalent either value may be returned. (5%/90%)

```
c:\>java Hotter
Enter F:>50.0
Enter C:>2.5
F is hotter!

Enter F:>-10
Enter C:>17
C is hotter!

Enter F:>32.0
Enter C:>0.0
C is hotter!

Enter F:>
```

```
import java.util.Scanner;
public class Hotter {
   static Scanner keyboard = new Scanner(System.in);

   private static double getDoubleFromUser(String prompt) {
      System.out.print(prompt);      // Prompt user to enter input value
      return keyboard.nextDouble(); // Read and return user's response
   }

   public static void main(String[] args) {
      double F,C;
      do {
         F = getDoubleFromUser("Enter F:>");
         C = getDoubleFromUser("Enter C:>");
         System.out.println(hotterOf(F,C) + " is hotter!");
         System.out.println();
      } while(true);
   }

   public static String hotterOf(double F, double C) {

   }

}
```

8. Consider the complete and correct Java program, named `Box`, on the left and an unfinished generalized program, `VarSizeBox`, on the right. `Box` simply draws a hollow box that is 5 characters wide by 5 lines long. `VarSizeBox` is designed to prompt for and accept the size of the box to be drawn as input.

As your answer to this question you are to provide the missing parts of `VarSizeBox` in the highlighted spaces. (10%/100%)

Fixed-Sized (5 by 5) Box

```java
public class Box {

   public static void main(String[] args) {
      drawBox();
   }

   public static void drawBox() {
      drawTop();
      drawBody();
      drawBottom();
   }

   public static void drawTop() {
      drawTopBottomLine();
   }

   public static void drawBody() {
      for(int i=0; i<3; i++) {
         drawBodyLine();
      }
   }

   public static void drawBodyLine() {
      System.out.print("|   |");
      System.out.println();

   }

   public static void drawBottom() {
      drawTopBottomLine();
   }

   public static void drawTopBottomLine() {
      System.out.print("+---+");
      System.out.println();

   }

}
```

Variable-Sized Box

```java
import java.util.Scanner;
public class VarSizeBox {
   static Scanner keyboard = new Scanner(System.in);
   static final char JOINT_SYMBOL = '+';
   static final char TB_SYMBOL = '-';
   static final char LR_SYMBOL = '|';
   static final char HOLLOW_SYMBOL = ' ';

   private static int getIntFromUser(String prompt) {
      System.out.print(prompt);
      return keyboard.nextInt();
   }

   public static void main(String[] args) {
      int dimension;
      dimension = getIntFromUser("Enter dimension:>");
      drawBox(dimension);
   }

   public static void drawBox(                    ){

      drawTop(                    );

      drawBody(                    );

      drawBottom(                    );
   }

   public static void drawTop(                    ) {

      drawTopBottomLine(                    );
   }

   public static void drawBody(                    ) {

      for(                                        ) {

         drawBodyLine(                              );
      }
   }

   public static void drawBodyLine(                    ) {

      System.out.println();
   }

   public static void drawBottom(                    ) {

      drawTopBottomLine(                    );
   }

   public static void drawTopBottomLine(                    ) {

      System.out.println();
   }
}
```